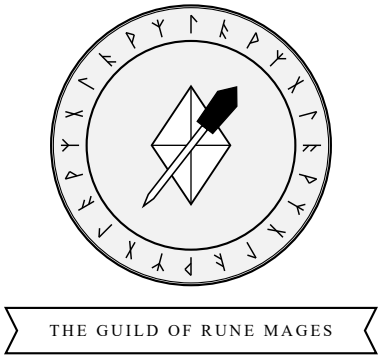
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SIGNATURES OF TRACES

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Signatures of Traces

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Abstract

A casting trace bears the mana pattern of its caster, and adjudication and checking have always known the caster by the pattern; but a higher law can imitate the pattern of a lower one, and the pattern alone cannot tell the true from the false. The two other methods now in use, shared-key check-marks and witnesses with sealing wax, each have their faults: whoever can check a mark can make it, or the witness cannot be present. This paper proposes trace signatures. Every caster and every calculating engine holds a private key, and enters a public seal in the register of public seals; when a trace is formed, a seal-box takes the digest of the trace, reckons a signature from it with the private key, and attaches the signature to the trace. Anyone can check a signature by the public seal, and every road to forgery analysed here requires first solving a question of arithmetic. The signature uses arithmetic alone and owes nothing to law; its safety rests upon an irreversible reckoning, in which the power-remainder is easy to reckon and the index is hard to find. By the fastest method of finding the index now known, all the engines of the League reckoning jointly would need a hundred thousand million years to recover the private key from a public seal of a hundred and fifty figures. The results of engines in other cities, signed layer by layer at each place, can be traced back to their input, to the version of the formula, and to the records of their checking. A signature proves from whose seal-box a trace came and which version of the formula it records; whether the content of the trace is true must still be examined apart. Following the custom of this journal, the universal units for signing and checking are appended.

1 Introduction

The casting trace is the first-rank evidence of adjudication. About C.R. 1130, after a wrongful judgment in a case of forbidden arts, Houghton revised the rules of evidence so that whenever an act of casting is to be established the trace is looked at first [1]; the trace is trusted because it bears the mana pattern peculiar to its caster. The acceptance of work in the workshops, the records of experiments, the inputs and outputs of calculating engines are all kept as traces in the same manner, for checking elsewhere [2].

Two things have lately made the pattern alone insufficient.

The first is that patterns can be imitated. About C.R. 1150 two limits were derived from nomological algebra, one of which says that a higher law can imitate a lower law, but not the reverse [3]. A pattern is one of the ways in which a law shows itself; a caster commanding

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a law higher than another's can therefore cast a trace bearing the other's pattern. This has long been known; but in former times very few could do it, and the courts had witnesses and material evidence to compare. Now that casting passes more and more through engines, and traces are more and more removed from the scene, the cases in which a trace is accepted on its pattern alone have multiplied. Of the false traces sent for re-examination by the examiners of Kanaich in the last twenty years, eleven in all, seven were made by imitating another's pattern with a higher law, and in two of these judgment had already been given. These are only the cases sent for re-examination, and cannot measure how many false traces there are; but they show where the trouble lies.

The second is that reckoning has left the hands of men. Since Fenn built the speaking network [4], a great reckoning is often divided among the engines of several cities, and the results are sent back over the network [2]. The man who wants the reckoning cannot see the engine; whether the numbers that come back were reckoned by that engine from the input given and by the formula given, he can only take on trust from the engine room of that city, and from everyone who relayed them on the way. Cummings has shown that no rune chain can judge the outcome of every rune chain [5]; a finished reckoning can of course be reckoned again or its record checked step by step, and restricted formulae can be verified in the manner followed since Finley, but all of this requires first knowing by whom, from what, and how it was reckoned.

Both things want the same: a mark which anyone can check to have come from a given person, and which no one but that person can make; and its trustworthiness must not depend on the rank of laws. The trace signature proposed here is such a mark. It uses arithmetic alone: no law, however high, makes a question of arithmetic easier to reckon.

Section 2 describes the three methods now in use and their faults; §3 the five conditions a signature must meet; §4 the irreversible reckoning on which it rests; §§5–6 the signature and the digest of a trace; §7 the seal-box, the register of public seals, revocation and attestation; §8 the chain of signatures for engines in other cities; §9 estimates the cost and the strength; §10 the relation of signatures to the Sixth of the Seven Millennial Problems; §11 what a signature does not prove; §12 concludes.

2 The three methods now in use

2.1 Examination of patterns The examiner compares the pattern of a trace, point by point, with the registered records of formulae and with the samples the caster left at the register [1]. This is the foundation of adjudication as it stands, and is very reliable against ordinary casters; against a forger commanding a higher law it cannot discriminate, as was said in §1. It also needs an examiner to do it with his own hands, and the outputs of engines and the records sent back over the network mostly have no pattern to compare.

2.2 Shared-key check-marks Rune messages on the speaking network are now mostly protected by shared-key check-marks. The two parties agree beforehand on a secret string of numbers, the shared key; the sender compresses the message together with the shared key into a check-mark, and appends it; the receiver compresses again in the same way, and if the two check-marks are alike he knows that the message has not been altered, and that it came from a holder of the shared key. Check-marks have two faults. First, whoever can check can make: the receiver holds the shared key too, and can make the same check-mark himself, so that a check-mark can make the two parties trust each other, but cannot be laid before

TABLE 1. The three methods and signatures

Requirement	*Patterns*	*Check-marks*	*Witness and wax*	*Signatures*
Anyone can check	needs an examiner	only holders of the key	must trust the witness	yes
Only the owner can make	a higher law can too	every holder of the key can	the seal can be imitated	yes, by the analysis of §5
No secret sent beforehand	yes	no	yes	yes
Quick to check	slow	quick	slow	quick
Independent of the rank of laws	dependent	independent	independent	independent

a third party as evidence. To use it in adjudication a court would have to agree a shared key with every caster, and anyone in the court who could reach the key could make that caster's check-mark. Secondly, the shared key must be delivered beforehand: if the parties have never met, the key must be carried by hand, since sent over the network it may be intercepted. The League has some millions of casters and the courts and engine rooms of some tens of cities; to agree keys pair by pair, the number of keys grows as the square of the number of persons.

2.3 Witnesses and sealing wax For important castings a witness is asked to be present, and afterwards signs the record and presses a seal in wax. Seals in wax can be imitated, and witnesses cannot stand before every calculating engine.

3 Five conditions

A signature must therefore meet five conditions:

(i) anyone can check it from public things alone, holding no secret;

(ii) no one without the private key can make it, and this holds also for those who can check it;

(iii) signer and checker need not meet beforehand, nor send each other any secret;

(iv) signing and checking are both quick: for a short trace, a fraction of a second on one way of an ordinary engine, with the work of compression added for a long one (§9);

(v) its trustworthiness rests on arithmetic alone, not on the rank of laws, nor on the honesty of the checker.

The first and second together are exactly what shared-key check-marks cannot do: in a check-mark, the secret used for checking is the same as the secret used for making. A signature must separate the two, signing with a private key and checking with a public seal, while the private key cannot be got from the public seal. The next section shows how this can be done.

4 Irreversible reckonings

4.1 Definition Take as the measure of size the number n of figures of p. Let the public numbers grow with n, and let the private key be taken uniformly at random within a prescribed range. A reckoning is called irreversible when the number of steps needed to reckon it forward grows as some power of n, while, for a result reckoned from a randomly

taken key, the number of steps needed by the best method now known to find any number that gives the same result grows faster than every power. In this sense irreversibility is only relative to the methods now known; it is an estimate of the cost of the existing methods, not a lower bound below which no method can go. No one has yet proved any reckoning to be truly irreversible; this bears on the Sixth of the Seven Millennial Problems, of which more in §10.

4.2 The power-remainder Take a large prime p such that $q = (p-1)/2$ is also prime, and a number g, not one, with $g^q \equiv 1 (\mathrm{mod}\, p)$, so that the powers of g leave exactly q different remainders on division by p. Whether p and q are prime, and whether g meets these conditions, is checked before witnesses when the register of public seals is opened. For x between 1 and $q-1$, put

$$y = g^x \bmod p \tag{1}$$

To reckon y from x is called taking the power-remainder. It is quick: write x in binary figures and, from the highest figure down, at each figure first multiply the number in hand by itself and, if the figure is one, multiply once more by g, keeping each time only the remainder on division by p; at most twice as many multiplications are needed as x has binary figures, and one and a half times on the average. To find x again from y is called finding the index. By the methods now known it is far harder:

(a) to try every value takes q steps;

(b) the method of great and small steps of Hollis [6]: take k a whole number not less than $\sqrt{q}$, first store the small steps $g^0, g^1, ..., g^{k-1}$, then, starting from y, multiply step by step by the great step g^{-k} until one meets a stored small step; both steps and storage are about $\sqrt{q}$;

(c) the method of a base of small primes of Spooner [7]: since the small primes need not be powers of g, one first takes a generator h of the whole group of remainders, of which g is the square; first, many relations among remainders involving only small primes are gathered; secondly, a set of linear equations modulo q is solved, giving the indices of these small primes to the base h; thirdly, y is resolved upon the small primes, its index to the base h found, and this divided by two modulo q gives the index to the base g. The steps needed are about

$$\exp\Big(\sqrt{2 \ln p \cdot \ln \ln p}\Big) \tag{2}$$

As the figures of p grow, (2) grows faster than every power, yet far more slowly than the $\sqrt{q}$ of (b); it is the fastest now known.

If the order of g has only small prime factors, the method of division of Salter [8] breaks the finding of the index into many small findings and is soon done. To avoid it the order of g must be a large prime; taking $p = 2q + 1$ is one way of meeting this condition, and it also makes possible the replacement of §6.

5 Public seals and private keys

5.1 Signing and checking Let the public p, q, g and f (f is described in §6) be fixed. Each signer takes a random number x between 1 and $q-1$, called his private key, and reckons y by (1), called his public seal. The private key exists only in the signer's seal-box; the public seal is entered in the register of public seals (§7).

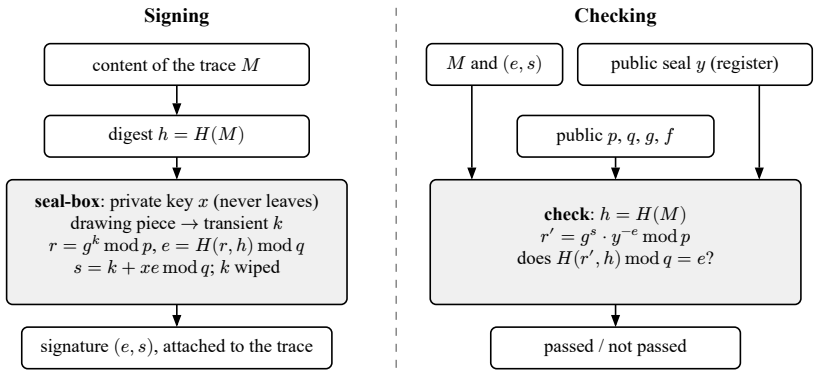


FIG. 1. Signing and checking: on the left, signing, the private key being only in the seal-box; on the right, checking, which uses public things only.

To sign a trace whose content is M, first reckon its digest $h = H(M)$ by §6. Then:

1. take a new random number k between 1 and $q - 1$, called the transient number;
2. reckon $r = g^k \bmod p$;
3. reckon $e = H(r, h) \bmod q$, that is, fill r out with noughts in front to the number of figures of p, set it before h, take the digest, and reduce it modulo q; call it the answering number;
4. reckon $s = k + xe \bmod q$, called the signing number.

The signature is the pair of numbers (e, s), and is attached to the trace (Fig. 1).

To check it one needs only the public p, q, g and f, the signer's public seal y, the content M of the trace and the signature (e, s). First see that e and s both lie between 0 and $q - 1$, and that y lies between 2 and $p - 1$ and $y^q \equiv 1 (\bmod p)$; if any of these fails, the signature does not pass. Then reckon $h = H(M)$ and

$$r' = g^s \cdot y^{-e} \bmod p \tag{3}$$

If $H(r', h) \bmod q = e$, the signature has passed.

LEMMA 1. *A signature made as in §5.1 always passes.*

Proof. The exponents are reduced modulo q, and the powers and products taken modulo p. $g^s y^{-e} = g^{k+xe} g^{-xe} = g^k = r$, since the q-th power of g leaves the remainder one; so $r' = r$, and the answering numbers on the two sides are the same.

Compression apart, to sign once takes one power-remainder; to check once, two.

5.2 Who can make one One without the private key who would make (e, s) for a new trace has three kinds of means, so far as analysed here. The first is to find the private key x, that is, to find the index of y. The second is to find two different contents with the same digest, and move the signature of one, which someone has signed, onto the other; Section 6 proves that this too comes down to finding an index. The third is to take e and s at will, reckon r', and see whether $H(r', h)$ happens to equal e; so long as the output of the digest follows no rule, this is mere trial, needing some q trials on the average. A forger may also see many genuine signatures, and even trick the signer into signing contents of his own choosing; whether signatures remain trustworthy in these cases I have not proved, and can say only that no faster means is known to me. The safety of signatures is offered here as a conjecture to be tested; the hundred thousand million years of §9 are only an estimate

for recovering the private key by the methods listed, not a lower bound for every means of forgery.

5.3 The transient number The transient number must not be used twice, nor be guessable.

Lemma 2. *If one private key with one transient number has signed two traces, with answering numbers not congruent modulo q, $e_1 \neq e_2$, then*

$$x = (s_1 - s_2)/(e_1 - e_2) \bmod q \tag{4}$$

Proof. $s_1 - s_2 \equiv x(e_1 - e_2) (\mathrm{mod}\, q)$; q is prime and $e_1 - e_2$ is not nought modulo q, so that one may divide.

The transient number must therefore be drawn on the spot by the drawing piece within the seal-box, and thrown away when used, neither stored nor recorded. The drawing piece reads the trembling of a fine crystal under mana, taking only the last few figures of each reading; readings are taken one after another until there are more than twice as many figures as q has, and the number is then reduced modulo q, so that the bias is too small to count. We took a hundred million numbers from each of ten drawing pieces of this Institute and tested them for bias in several groupings and for dependence between neighbours, at ordinary warmth, after restarting, and with casting nearby, and could find no fault; the tests and their statistics are given in the supplement. If the drawing piece gives no reading, or repeats one, the seal-box stops signing. Since the transient number is drawn at random, the chance that two of N signatures happen to share it is about $N^2/(2q)$, which at the figures of §9 may be neglected. If the transient number can be guessed, the private key can be solved for after the manner of (4).

5.4 A side remark: agreeing a shared key in the open The same reckoning has one more use, not treated at length here. Two persons who have never met take, the one a, the other b, and send g^a and g^b; the first raises the other's number to the power a, the second the other's to the power b, and both obtain g^{ab}, which can serve as their shared key. One who intercepts the two numbers can reckon g^{ab} if he can find an index; whether reckoning it must go through finding an index, no one has proved. This guards only against eavesdropping, not against a relayer who impersonates: he may replace both numbers with his own, and agree a separate key with each party. Against that, the numbers of both parties must be sent signed, together with their identities.

6 The digest of a trace

The content of a trace is long: the record of the pattern, the time, the place, the version of the formula used, the input and output of an engine. What is signed is its digest, a number of fixed length reckoned from the content. The digest must meet two conditions: from the digest no content agreeing with it can be found; nor can two different contents with the same digest be found. The latter is called freedom from collision.

We compress by the power-remainder. Besides g, take a public number f lying in the group of remainders generated by g, but whose index to the base g no one has handed over: on the day the register is opened, numbers are cast before witnesses to give a remainder u, not nought, and $f = u^2 \bmod p$ is taken, the cast being repeated if f is one; the whole cast is written at the head of the register. This shows only that no index appeared in the fixing of f, not that no one knows it.

The content is written in decimal figures and divided from the beginning into pieces of m figures, m being taken so that 10^m is less than q; the last piece, if short, is filled out with noughts in front, and one further piece records the total number of figures of the content, which therefore may not reach 10^m. Every piece then has a value between 0 and $q-1$; call them $M_1, M_2, ..., M_n$. Put $c_0 = 1$ and

$$c_i = g^{c_{i-1}} \cdot f^{M_i} \bmod p \tag{5}$$

where c_{i-1}, if greater than q, is first replaced by $p - c_{i-1}$. Every c lies in the group generated by g; since q is odd, minus one is not in that group, so that c and $p-c$ are never both in it, the numbers so replaced never exceed q, and different c give different numbers. The last c_n, so replaced, is the digest.

LEMMA 3. *Whoever finds two different contents with the same digest can find the index of f to the base g.*

Proof. Let the two contents be compressed piece by piece into c_i and c'_i. The last pieces record the total numbers of figures, so that if these differ the last pieces differ. Since the digests agree and the replacement is one-to-one, the last c of the two chains agree; going back step by step from the end of each chain, find the first place where the c agree but $(c_{i-1}, M_i) \neq (c'_{j-1}, M'_j)$; since the contents differ, there is such a place. Let a and a' be c_{i-1} and c'_{j-1} after replacement, so that $g^a f^{M_i} = g^{a'} f^{M'_j}$. If $M_i = M'_j$, then $g^a = g^{a'}$; a and a' do not exceed q, so they are equal, contrary to supposition. Hence $M_i \neq M'_j$; both lie between 0 and $q-1$, so that their difference is not nought modulo q, and the index of f to the base g is $(a - a')/(M'_j - M_i) \bmod q$.

Freedom from collision therefore also rests upon finding the index: whoever can make a collision of digests has found an index. Compression takes two power-remainders for each piece and is slower than signing. A compression of the pieces in parallel would be a different digest, not covered by Lemma 3, and is not used here.

7 The seal-box, the register of public seals, revocation and attestation

7.1 The seal-box The private key is kept in the seal-box and never leaves it. The seal-box is a small instrument carried on the person, containing the wafer that holds the private key, a drawing piece and a reckoning unit. For each signature the holder must open the seal-box with a pass-word; after three wrong attempts the seal-box locks, and the holder must go in person to the register office to have it reopened. When the caster casts, or an engine writes a record, the seal-box takes the digest of the version directly from the formula being cast or executed, through a sealed interface, records it together with the content of the trace, signs both, and notes in the signature "taken through the interface"; text passed into the seal-box at will is signed too, but without that note. The case of the seal-box carries a ward, which wipes out the private key if the box is forced. That ward guards against opening the box to read the key; one who steals the whole seal-box and knows the pass-word can sign falsely, so the pass-word must be kept in the head, not written on the box.

7.2 The register of public seals A public seal must answer to a person, or a forger need only make another pair of private key and public seal and register them in another's name, to sign "another's" signatures. The registration of a public seal must therefore be made in person, like the oath register: the signer brings his seal-box to the oath-register office; the registrar establishes his identity by the original roll of the oath register, its certificates, the

trace of their making and any later changes, the identity being determined by the registering authority according to law and not by pattern alone; he has him sign before him, with the seal-box, a content given by the office, checks it, and enters in the register of public seals the name, the oath-register number, the number of the public seal, the set of p, q, g and f used, the public seal, and the days from which and until which it holds; and the office then signs that entry with its own private key. The public seals of the register offices are printed each year in the Gazette of the League, and the original of the Gazette kept by the courts is the starting point of trust. Anyone who holds a trace, then, first checks the signature of the trace and then the signature of the entry in the register, and knows who signed. The seal of an engine is registered on its behalf by the body that keeps the engine room. A change of key is made by registering a new public seal, the old entry being marked with the day it ends; if a register office loses its own private key, it is revoked in the same way, and the office above it signs the new entry.

7.3 Revocation If a seal-box is lost, or the private key is suspected to have leaked, the holder asks the register office to revoke it; the office records the time and prints the revocation in the Gazette. Signatures made after revocation are not accepted. The checker must therefore obtain a sufficiently recent record of revocations, and may not rely on an out-of-date Gazette.

7.4 Attestation One who has stolen a seal-box might still write the time of a signature earlier than it was. Against this I propose two things. First, that the speaking network send a time signal every hour, itself signed, and that every trace record the latest time signal at the moment of signing; this shows only that the signature is no earlier than that time signal. Secondly, that the digest of a signed trace, together with its signature and the number of the public seal, be sent promptly to the time office, which signs a receipt stating the time at which it received them; this shows that the signature is no later than the time of the receipt. Together, the time of the signature is held between two times. Where these enclose the time of a revocation, the trace is examined apart, under the third head of §11.

8 The chain of signatures for engines in other cities

A reckoning in another city is signed thus (Fig. 2). The party A that wants it gives the reckoning a number, writes the number, the input I and the version V of the formula to be used into one content, signs it with his private key, and sends it to the engine room of city B. B's engine reckons the output O by V, and at the same time records K, the record of its checking — the reckonings made at two places, the checks used, the bounds of error; the records of the reckonings at the two places are signed by each place separately, and K only cites them, not reported by one place for the other. B then signs with its engine seal a content made by joining the number, I, V, O, K and A's signature. In a joint reckoning each part is signed in this way by the engine that reckoned it, noting which part it is and on which parts it depends; the engine room presiding over the joint reckoning signs the result it has combined, together with the signatures of the parts, and the final digest is sent for attestation as in §7.4.

A result so signed can be checked by anyone layer by layer: from which engine room the final result came; which partial results it combined; by which engine, from which input, by which version of the formula each was reckoned; what checks were made at each place; and who signed the first input. A signature does not warrant that a reckoning is right. Where the

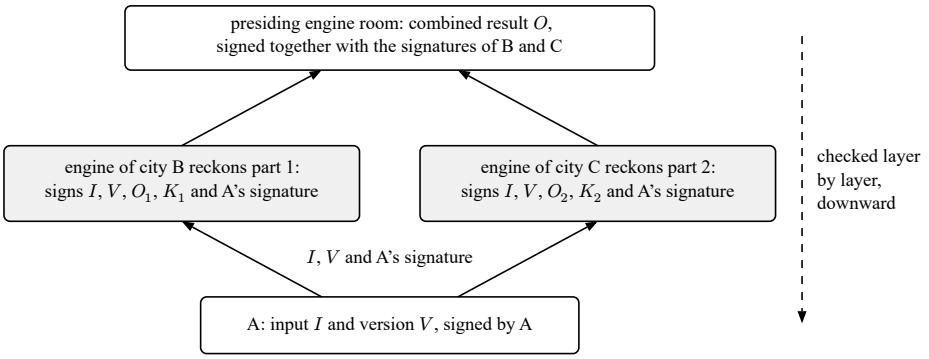


FIG. 2. The chain of signatures for engines in other cities: A signs the input; the engines of cities B and C each sign the part they reckoned; the engine room presiding over the joint reckoning signs the combined result; each layer joins the signatures of the layer below into the content it signs.

records are complete and can be reckoned again independently, if it is wrong, one can find at which place and by which version of the formula it went wrong; and no one can afterwards alter any one layer without its being detected. If one layer of signatures is missing, the chain as a whole does not stand, but the signatures of the layers below still prove their own sources.

9 Cost and strength

Table 2 gives, for primes of different numbers of figures, the time to sign and to check, and the time of the two methods of finding the index. Signing and checking are reckoned on one way of the engine room of this Institute, compression apart: the four rows from forty to a hundred figures were measured, each by a thousand signings and checkings averaged, and the rest extrapolated as the cube of the number of figures, since a multiplication with remainder grows as the square of the number of figures, and a power-remainder needs a number of such multiplications in proportion to it. Compression takes two power-remainders for each piece; a trace of a thousand figures, at a hundred and fifty, takes about a second and a half. Finding the index is reckoned on all the engines of the League reckoning jointly: by the count of the Seventh Division this year, the engines registered in all the cities of the League together make some two thousand four hundred million multiplications with remainder of a hundred figures every second. The two columns for finding the index assume ample storage and perfect parallel working, and count every step of (2) as one multiplication with remainder, neglecting its constant and giving only the order of magnitude; they are an ideal budget of work, not a lower bound of safety. In practice the method of great and small steps must also store some $\sqrt{q}$ numbers, and the step of solving the linear equations in the method of a base of small primes cannot be wholly divided among ways. There is a further point: when the whole League shares one set of p and g, the first two steps of the method of a base of small primes need be done only once, after which the private key of each public seal is found much faster; this is one reason for changing the set of p and g every twelve years.

From a hundred figures to two hundred, for example, signing and checking become some eight times slower, while finding the index by the method of a base of small primes becomes

Table 2. Figures of the prime, and the time to sign, to check, and to find the index

Figures of p	*To sign (compression apart)*	*To check (compression apart)*	*Great and small steps*	*Base of small primes*
40	0.002 s	0.003 s	150 years	4 minutes
60	0.005 s	0.011 s	3×10^{12} years	18 days
80	0.013 s	0.026 s	6×10^{22} years	90 years
100	0.025 s	0.050 s	10^{33} years	70,000 years
120	0.043 s	0.086 s	10^{43} years	30 million years
150	0.084 s	0.17 s	2×10^{58} years	10^{11} years
200	0.20 s	0.40 s	4×10^{83} years	2×10^{16} years

three hundred thousand million times slower; on this alone the possibility of signatures depends.

How many figures to take depends on how long a signature must hold. The power of calculating engines has grown some tenfold every ten years for the last thirty years, and the methods of finding the index may be improved. The Silver Folk are not counted old until two hundred and ten, and the traces of a case or of a contract may be brought out for re-examination after some hundreds of years. Were the power of engines to go on growing tenfold every ten years, the hundred thousand million years of budget at a hundred and fifty figures would be spent in about a century, and that at two hundred figures in about a century and a half; this is a supposition, not a prophecy, but it is enough to show that the matter cannot be settled once for all. I propose that ordinary traces take a hundred and fifty figures; that the register of public seals, the public seals of the register offices and the original records of trials in court take two hundred; that these be only a starting point, the First Division re-estimating every twelve years by the power and the methods of the time; and that, when the figures are about to become insufficient, the traces be signed afresh at the new figures while the old signatures can still be relied upon. The new signature must cover the original content, the old signatures, the entry in the register and the receipts of attestation, and the record of their checking at the time of re-signing must be kept. A signature added after the old one has become forgeable cannot restore the truth of what went before.

10 Signatures and the question of verification and construction

The Sixth of the Seven Millennial Problems asks whether, for a hard problem in which a rune chain, once given, can be quickly verified, the length of the chain growing only moderately with the size of the problem, such a chain can always be quickly constructed as well [9]. Forging a signature can be written as such a problem: a short signature, once given, is checked by two power-remainders; to forge, one must construct it. If the answer to the Sixth Problem were “yes”, with a practicable method of construction, signatures could no longer be kept safe. Conversely, even an answer of “no” would not by itself make the power-remainder irreversible: the Sixth Problem asks whether any problems of this kind are hard, while signatures need a randomly taken public seal to be hard against every effective means in the ordinary case. Among randomly taken private keys a few are weak — with $x = 1$, y is simply g — but their proportion may be neglected. The trustworthiness of signatures rests, in the end, only on what is “now known”. What we can do is to make it rest on a

question that many have tried and none has yet solved. The finding of the index has been attempted since the reckoners of the classical age [6].

11 What a signature does not prove

A signature proves that the content of a trace, at the time of signing, came from the seal-box holding the private key that answers to a certain public seal; where it bears the note "taken through the interface", it proves also that the version recorded in the content was taken by the seal-box from the formula being cast or executed. It does not prove the following.

(i) It does not prove the content true. A signer may cast a false formula and have it leave a true signature; whether the content of a trace is true, and how the act of casting and the responsibility for it are to be judged, remain matters for examination and trial.

(ii) It does not prove that the signer acted of his own will. A seal-box signs for a man under duress as well.

(iii) It does not prove that the private key was not stolen. In arithmetic, no law however high makes the index easier to find; but the ward of a seal-box, which keeps out ordinary casters, may not keep out a higher law, and one who has stolen a seal-box and knows its pass-word can sign falsely. That a seal has not been revoked does not mean that its key never leaked. The safety of the private key rests in the end on the seal-box never leaving the person and the pass-word never being told, and on revocation as soon as either is lost.

(iv) It does not reach back. Traces formed before signatures came into use have no signatures, and are still to be examined under the old rules, by the record of their collection, the course of their keeping and the findings of examination; they may not be refused merely for having no signature.

Checking a signature is only one step towards accepting a trace. The warrant for collection, the chain of keeping and cross-checking all remain as required since Houghton.

12 Conclusion

The marks by which men are known, from sealing wax to the pattern, have always been things hard to imitate: the wax wants the die, the pattern wants the man's law. What is hard to imitate can always be imitated by someone. A signature goes another way: it is not hard to write; what is hard is to reckon it without knowing the private key; and anyone can check it. To trust from whom a trace came and which version of the formula it records, then, one need no longer rely on the pattern alone being hard to imitate, nor on the honesty of everyone who relayed it on the network, but on a question of arithmetic that has not yet been solved, together with the registration of public seals, the keeping of seal-boxes and attestation. The rest of the examination must be made as before.

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References

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3. The White Tower Society, "The two limits of law", Annals of the White Tower Society (C.R. 1150).
4. Fenn, The speaking network, Tallin Polytechnic University, F.E. 191.

5. Cummings, “On decidable matters, with an answer to Severin’s Fifth Question”, Annals of the Laws 223 (F.E. 319).
6. Hollis, “The method of great and small steps for finding the index”, Annals of the Laws 112 (F.E. 208).
7. Spooner, “Finding the index on a base of small primes”, Annals of the Laws 228 (F.E. 324).
8. Salter, “Finding the index when the order has only small prime factors”, Annals of the Laws 215 (F.E. 311).
9. The Millennial Foundation, The Seven Millennial Problems, Kanaich, F.E. 307.
10. Ashby, Universal rune units, Lionwing, c. F.E. 200.

Appendix. The universal units for signing and checking

Following the custom of this journal, the universal units for signing, checking and compression are appended [10]. Numbers are kept as whole numbers, and all reckoning is exact whole-number reckoning: the intermediate product of a multiplication with remainder is kept to twice the figures of p, and cut back only after the remainder is taken; multiplication with remainder, squaring and the taking of inverses use the standard sub-chains of the supplement to this number. The supplement also gives sets of test numbers: a fixed private key, transient number and content, with the signature that should result; an empty content; a content whose last piece is filled out; a signature with answering number nought; an unlawful public seal; a content altered in one figure; and every case that ought to be refused. After engraving, the units should be compared with another set written independently, not merely with the output of the same set. ⟨Sign⟩ may be engraved only within a seal-box, the readings of the drawing piece passing straight into the unit, not through the store.

```
-- power-remainder: the remainder of b to the power x on division by p;
-- the binary figures of x are read from the highest
⟨PowerRem b, x, p⟩
  c ← 1
  for each binary figure t of x, highest first:
    c ← c · c mod p
    if t = 1: c ← c · b mod p
  give c
-- compression: the content in pieces as in §6, the last recording
-- the total figures; fold(c) is p − c when c exceeds q, else c
⟨Compress M⟩
  c ← 1
  for each piece M[i]:
    c ← ⟨PowerRem g, fold(c), p⟩ · ⟨PowerRem f, M[i], p⟩ mod p
  give fold(c)
-- signing: within the seal-box only; opened by pass-word;
-- k drawn on the spot by the drawing piece, discarded when used
⟨Sign M⟩
  h ← ⟨Compress M⟩
  k ← drawing piece (1 to q − 1); stop if no reading or a repeated one
  r ← ⟨PowerRem g, k, p⟩
  e ← ⟨Compress r joined to h⟩ mod q
  s ← (k + x · e) mod q
  wipe k
  give (e, s)
-- checking: public p, q, g, f and the public seal y only
⟨Check M, e, s, y⟩
  if e or s lies outside 0 to q − 1: give “not passed”
  if y lies outside 2 to p − 1: give “not passed”
  if ⟨PowerRem y, q, p⟩ ≠ 1: give “not passed”
  h ← ⟨Compress M⟩
  r′ ← ⟨PowerRem g, s, p⟩ · ⟨PowerRem y, q − e, p⟩ mod p
  if ⟨Compress r′ joined to h⟩ mod q = e: give “passed”
  else: give “not passed”
```